

LLM-Augmented High-Level Route Planning for Autonomous UAV Navigation: A Survey

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Short Survey

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1 Scope

This survey examines the use of large language models (LLMs) and vision-language models (VLMs) as high-level route planners for autonomous UAV navigation, with particular focus on architectures where the foundation model contributes semantic reasoning and contextual interpretation while classical algorithms retain responsibility for geometric refinement, trajectory feasibility, and low-level control. The survey is **bounded to works that couple FM-based planning with real or simulated UAV execution** and excludes purely ground-robot navigation and general embodied AI benchmarks. The central question addressed is whether semantic reasoning and generalisability across various environments, obstacles and robotic platforms capabilities of these foundational models can be leveraged to generate safe, dynamically feasible and above all real time deployable plans for UAVS and what are some potential gaps and tradeoffs one has to encounter.

2 Main Approaches

The literature naturally organizes into four clusters defined by the role the foundation model plays relative to the underlying planner.

2.1 Cluster 1: End-to-End FM Planners

The earliest and most direct formulation treats the FM as the complete planner. The system receives start, goal, and obstacle information as a structured text prompt and outputs a waypoint sequence directly.

FM-Planner [Xiao et al., 2025] provides the most comprehensive study of this approach in the drone domain. It benchmarks eight LLM and five VLM configurations across simulated indoor scenarios and extends to real-world hardware experiments. In its first stage, an LLM such as Llama-3.1-8B-Instruct processes start, goal, and obstacle coordinates serialized as natural language tokens and generates an initial coarse trajectory. The paper reports that Llama-3.1-8B-Instruct matches A* in success rate on two- and three-obstacle scenarios while maintaining 100 percent success rate up to five obstacles, whereas competing models including Qwen-2.5-coder-32B and all Gemma variants degrade substantially with increasing obstacle count. Average reasoning time is approximately 9 seconds per query. The paper explicitly notes this latency is acceptable because global path plans do not require high-frequency updates. The VLM-guided variants, which receive annotated visual map images rather than text coordinates, consistently produce less reliable and less optimal paths than text-coordinate LLMs, suggesting that the textual representation of spatial information is better-suited to FM reasoning than raw visual

input.

Navigating Motion Agents in Dynamic and Cluttered Environments [Zhao et al., 2025] extends this cluster to multi-agent settings and continuous-space environments. Rather than restricting movement to four-directional grids as in prior works, it encodes floor plans and obstacle positions as discrete text tokens and generates anchor-point trajectories — sparse sequences of key positions connected by straight segments. Using o3-mini, single-agent success rate reaches approximately 80 percent in a single trial and approximately 90 percent with multi-turn refinement on realistic floor plans. The paper also demonstrates closed-loop replanning where agents query the LLM when a planned segment is blocked, enabling dynamic obstacle handling without retraining.

Exploring Spatial Representation to Enhance LLM Reasoning in Aerial VLN [Gao et al., 2025] addresses a structural weakness of end-to-end FM planners directly. It observes that existing LLM-based VLN methods describing visual observations in free-form text or maintaining topological graphs are insufficient for aerial scenes where large-scale spatial relationships must be precisely encoded. To address this, it proposes the Semantic-Topo-Metric Representation (STMR): depth images and segmentation masks are projected into a growing top-down map, which is discretized into a 20×20 matrix with semantic labels and distance metrics. This matrix is provided as the text prompt to the LLM rather than raw images or coordinate lists. The paper demonstrates that direct visual input to VLMs — even GPT-4o — produces dramatically worse navigation results than the matrix representation, confirming that structured spatial encoding is critical for reliable LLM spatial reasoning. On real UAV hardware using a Q250 airframe with an Intel RealSense D435i, the method achieves 40 percent success rate across 10 outdoor scenes versus 20 percent for the best competing VLM baseline. This work is directly relevant as it shows that bridging the gap between raw sensor data and LLM-interpretable representations substantially improves planning reliability.

What is demonstrated: Success rate improvements over baselines in simulation for sparse obstacle environments. **The main limitation is that end-to-end FM planners hallucinate spatially infeasible paths in dense obstacle configurations, and their geometric precision degrades without external verification.**

2.2 Cluster 2: FM Augmenting Classical Search

Rather than replacing the planner, **these works inject FM-derived semantic information into the cost functions, sampling distributions, heuristics, or parameter configurations of classical algorithms.**

SRAH [Durrani and Durrani, 2026] encodes LLM-inspired spatial risk reasoning directly into an A* cost function without requiring LLM inference at deployment time. Cells surrounded by three or more blocked neighbors receive a penalty of 2.0; those adjacent to two blocked cells receive 0.8; open cells receive no penalty. This operationalizes the LLM principle that geometrically constrained zones with limited escape routes carry higher operational risk. Evaluated across 200 trials in a 15×15 grid with 20 percent static obstacle density and stochastic dynamic obstacles, SRAH achieves 62.0 percent task success versus 56.5 percent for BFS with replanning and 4.0 percent for greedy without replanning. Planning overhead is 2.61ms per query — orders of magnitude below LLM inference latency. The paper explicitly acknowledges that the current evaluation is confined to discrete grid worlds and that the cost function relies solely on static geometry without object semantics or agent velocity.

VLM-RRT [VLM-RRT, 2025] incorporates a VLM module into the RRT sampling loop to analyze environmental snapshots and bias sampling toward regions more likely to contain feasible paths. A parameter γ controls the strength of VLM-guided sampling bias. The paper demonstrates via ablation that $\gamma = 1$ causes greedy behavior and local minima, while $\gamma = 0$ degrades to standard RRT — finding the appropriate balance remains an open tuning problem explicitly noted by the authors.

LLM-DWA [LLM-DWA, 2025] separates semantic waypoint generation from reactive obstacle avoidance. An LLM produces intermediate navigation targets based on contextual scene understanding, while the Dynamic Window Approach handles local obstacle avoidance at runtime. The paper demonstrates that incorporating LLM-generated waypoints enhances DWA’s navigation success rate and reliability in environments where purely reactive control would otherwise produce suboptimal or failed trajectories.

Two closely related works extend this cluster to the **domain of policy synthesis and parameter optimization through code generation**. Bosio et al. [Bosio et al., 2025a] propose combining LLMs with gradient-free optimization in a human-in-the-loop architecture for automatic controller synthesis. The LLM generates structured controller code from high-level task descriptions, but rather than treating the generated output as final, the framework invokes gradient-free optimization routines to compute optimal parameter configurations for the generated controllers. The optimized parameters are then provided back to the LLM, which refines its code in response — effectively using the LLM as a semantic scaffolding layer around an optimization backbone rather than as a numerical parameter estimator. This augmentation pattern directly addresses the **known weakness of LLMs in producing numerically precise parameter values** while retaining their advantage in generating interpretable, semantically grounded policy structures.

A companion paper [Bosio et al., 2025b] extends this approach toward synthesizing interpretable control policies through LLM-guided program synthesis, where the LLM’s role is to propose program structures that encode meaningful behavioral abstractions, with correctness and performance verified by external computational tools. Together, these works establish the principle that LLMs used for code or policy generation — rather than direct numerical output — combined with tools that handle the numerical optimization burden, represents a robust and practically deployable architecture.

SafePath [SafePath, 2025] offers a conceptually different simplification strategy. **Rather than generating numerical waypoint coordinates, it formulates navigation as a discrete action selection problem** where the LLM selects from a predefined vocabulary of high-level actions expressed in natural language — for example, *turn left, go straight, slow down*. By **eliminating the need for the LLM to produce spatial coordinates entirely**, SafePath **bypasses the numerical precision failure mode** that affects coordinate-generating planners. Conformal prediction is employed to provide probabilistic safety guarantees over the action distribution, enabling formal bounds on constraint violation rates. However, this simplification introduces a fundamental tradeoff: by restricting the LLM to a discrete action vocabulary, the framework **sacrifices the rich semantic and contextual generalizability that makes FM-based planning appealing** in the first place. .

Take home message: The key gap is that none of these systems have been evaluated at scale in **unstructured real-world environments**, and the **VLM sampling bias parameter in VLM-RRT** remains a manually tuned hyperparameter rather than an adaptively learned one.

2.3 Cluster 3: Deployment-Oriented and Lightweight FM Planners

These works prioritize real-world deployability, hardware constraints, and the use of small FM models running on resource-constrained platforms.

LLM-Land [Cai et al., 2025] addresses autonomous drone landing in dynamic environments using a hybrid framework combining a vision-language encoder, a lightweight RAG-augmented LLM, and model predictive control. **A BLIP encoder converts onboard camera images to scene captions, which are processed by a quantized Llama-3.2-1B or Qwen-2.5-1.5B model to infer semantic obstacle classifications and context-aware safety buffers** — for example, 3 meters of clearance for pedestrians and 5 meters for vehicles. The RAG mechanism retrieves relevant entries from an onboard knowledge base to stabilize LLM outputs and prevent hallucinations. The MPC module uses the LLM-inferred constraints to replan collision-free landing trajectories in real time.

Tested across 50 trials in a Gazebo open-field environment, LLM-Land achieves a 96 percent success rate versus 34 percent for baseline MPC and 6 percent for LLM-without-RAG. The paper explicitly attributes the **catastrophic failure of the LLM-without-RAG variant to semantic hallucinations** that disrupt rather than improve MPC control — a finding that directly motivates the verification-augmented prompting explored in the present project. A hardware benchmark across quantized models shows that Llama-3.2-1B-Instruct delivers the optimal tradeoff at 1453ms latency, 1.5GB VRAM, and 94 percent success rate on a Jetson platform.

ICL for Public Safety UAV [Emami et al., 2026] proposes **deploying LLMs at network edge servers rather than cloud infrastructure, eliminating latency** and privacy concerns for time-sensitive UAV operations. It formalizes a unified framework where LLM-assisted in-context learning supports path planning, velocity control, data collection scheduling, and power management through **structured task descriptions** comprising task goal, input data, rules, constraints, output specification, and a feedback loop. **ICL is inherently lightweight because it eliminates retraining entirely**, adapting to new environments **purely through prompt design and contextual demonstrations**. A comparison table shows ICL requires only 1 training episode of 30 steps and approximately 20 seconds on CPU versus 1,000 episodes and 50 GPU-minutes for DRL. Inference latency is reported at 0.3–0.7ms average. The framework demonstrates the generalizability advantage of LLMs: the same model can be repurposed across path planning, velocity control, and power management simply by modifying the task description, without any parameter updates.

LEMUR [Dong et al., 2025] validates LLM-FM integration on physical DJI Tello hardware using Gemini 2.5 Flash as a mission planner. The system exposes 18 tool functions wrapping low-level SDK primitives and evaluates performance across obstacle navigation, search and rescue, and return-to-base phases. Results demonstrate that **while the state machine completes obstacle traversal twice as fast due to its continuous feedback loop, the LLM agent achieves faster mean end-to-end mission completion** across the full three-phase scenario, demonstrating that mission-level performance is not simply the sum of individual phase characteristics. The **paper identifies planning latency as the primary cost**, with the agent spending approximately 40 percent of obstacle traversal time stationary while awaiting LLM inference. This finding **confirms that LLM-based planning overhead must be absorbed at the mission level rather than requiring real-time operation**.

Take home message: The main limitation is that all hardware demonstrations operate in **relatively controlled indoor environments** with limited obstacle diversity with

latency being primarily challenge these papers aim to bypass.

2.4 Cluster 4: High-Level Context-Aware Decomposition

These works focus on the LLM as a **semantic task decomposer that produces mission-level subgoals**, leaving all navigation to downstream classical systems.

DCIP [Doma et al., 2025] positions the LLM as an occupancy grid manipulator rather than a waypoint generator. Natural language instructions such as *avoid the repair area* or *prefer empty lanes* are parsed by the LLM into structured cost-modification actions that update traversal costs in specific grid regions. A downstream A* planner searches the modified grid. Experiments with Llama3, Llama3.1, and Mistral on a Husky-UR3 robot in a Gazebo industrial warehouse demonstrate that Llama3 achieves 92.85 percent instruction adherence and 96.42 percent safety compliance. DCIP expands significantly fewer nodes than standard A* in larger environments because the LLM-modified cost landscape steers search away from semantically undesirable regions early in the process.

The **STMR** paper [Gao et al., 2025] also contributes to this cluster by maintaining a state machine of subgoal completion statuses — *todo*, *in-process*, and *completed* — which prevents the LLM from re-attempting completed subgoals and reduces hallucination-driven replanning. Ablation results show that subgoal state tracking increases success rate from 9.0 percent to 15.0 percent on complex navigation benchmarks, attributable entirely to reduced context inconsistency hallucination.

Take home: LLMs demonstrated **Zero-shot instruction following** in simulation and limited real-world settings. The primary limitation is that **subgoal decomposition remains decoupled from geometric feasibility** — the LLM may decompose a task into subgoals that are individually feasible but collectively inconsistent.

3 What Has Actually Been Shown

The evidence base across this literature is heterogeneous and requires careful interpretation.

FM-Planner [Xiao et al., 2025] provides the most systematic benchmarking, evaluating eight LLMs and five VLMs across controlled simulation scenarios with quantitative success rate, path length, and timing metrics, followed by real-world validation on physical hardware. This is the strongest evidence base in the surveyed literature.

The STMR paper [Gao et al., 2025] validates on both simulation benchmarks and ten real outdoor scenes with ground-truth trajectory measurement, providing meaningful evidence that structured spatial representations generalize beyond simulation.

LLM-Land [Cai et al., 2025] provides statistically grounded results across 50 trials per condition with clear baselines, making it one of the more rigorous hardware evaluations in the deployment-oriented cluster.

LEMUR [Dong et al., 2025] provides real hardware timing data but across only three trials per condition, limiting statistical interpretability.

SRAH [Durrani and Durrani, 2026] provides 200 randomized trial comparisons across multiple planners and obstacle densities — the most statistically robust evaluation in the FM-augments-classical cluster.

Several papers including DCIP [Doma et al., 2025], the ICL UAV paper [Emami et al., 2026], and the Bosio works [Bosio et al., 2025a,b] rely primarily on qualitative demonstrations or limited trial counts where reported improvements are plausible but cannot be generalized to diverse environments with confidence.

A recurring pattern across the literature is that results demonstrated on sparse-obstacle structured environments **do not consistently transfer to dense, cluttered, or dynamically changing environments. Performance gaps between simulation and hardware are noted explicitly** in FM-Planner [Xiao et al., 2025] and LEMUR [Dong et al., 2025] and implied in most other works.

4 Open Problems and Limitations

Spatial hallucination and geometric imprecision. The most fundamental limitation across all clusters is that LLMs generate geometrically inconsistent outputs — waypoints that collide with obstacles, violate spacing constraints, or produce non-monotonic progress toward goals — without any internal mechanism to detect or correct these errors. The STMR paper [Gao et al., 2025] identifies this as a structural limitation distinct from poor prompting, attributing it to the inability of LLMs to maintain spatially consistent internal representations across long reasoning chains. The Navigating Motion Agents paper [Zhao et al., 2025] confirms this finding: even in simple floor plan environments, models frequently plan paths that cross directly through obstacles via the most direct route. This is **not a prompting problem that better instructions fully resolve — it is a structural property of how LLMs represent geometric space as sequences of tokens rather than as metric spatial structures.**

Latency incompatibility with reactive control. FM inference latency of 9 seconds per query as reported in FM-Planner [Xiao et al., 2025], and 300–600ms per tool call in LEMUR [Dong et al., 2025], is fundamentally incompatible with reactive obstacle

avoidance at control rates of 10–50Hz. LEMUR quantifies the cost directly: the LLM agent spends 40 percent of obstacle traversal time stationary awaiting inference. Current architectures address this by separating high-level planning from low-level execution, but this separation limits the system’s ability to respond to dynamic obstacles encountered during trajectory execution.

Absence of formal safety verification. No surveyed paper provides formal guarantees that LLM-generated trajectories are collision-free prior to execution. **Most systems rely on downstream collision checking or reactive avoidance as safety mechanisms.** SafePath [SafePath, 2025] represents the closest attempt to formal safety guarantees through conformal prediction over discrete action distributions, but achieves this by abandoning numerical coordinate generation entirely. The SRAH paper [Durrani and Durrani, 2026] provides probabilistic improvements but not formal guarantees. LLM-Land [Cai et al., 2025] introduces the closest analog to a verification layer through the RAG-grounded safety buffer mechanism, but this operates at the level of constraint specification rather than trajectory verification. Using libraries like **Auto Lipra** [?] that claim to provide formal linear bounds on outputs of distribution free black box models including natural language transformers can be an interesting research direction one can take.

Limited diversity of real-world validation. Across the surveyed papers, real-world hardware experiments are conducted in controlled indoor environments — FM-Planner in a lab space with cubic foam obstacles, LEMUR in a room with standardized pole configurations, LLM-Land in Gazebo with real hardware landing tests, and the STMR paper in ten curated outdoor scenes. None of the papers demonstrate robust performance across diverse unstructured environments with varying obstacle density, dynamic objects, and lighting conditions.

Dependence on structured environmental representations. The STMR paper [Gao et al., 2025] demonstrates that raw visual inputs to VLMs produce dramatically worse navigation performance than structured matrix representations. This means reliable FM planning currently requires non-trivial **intermediate processing — segmentation, projection, discretization — that introduces its own failure modes and latency.** A general-purpose FM planner that reliably interprets raw sensor data without intermediate structuring remains beyond the current state of the art.

Numerical precision versus semantic generalizability tradeoff. SafePath [SafePath, 2025] and the Bosio et al. works [Bosio et al., 2025a,b] collectively expose a fundamental tradeoff that the field has not yet resolved. SafePath achieves safety by eliminating coordi-

nate generation entirely, sacrificing the spatial reasoning capabilities that distinguish FM planners from classical systems. The Bosio approach preserves semantic expressiveness by generating policy code rather than numerical parameters, delegating numerical computation to external optimization tools. Both strategies concede that LLMs cannot reliably produce numerically precise spatial values on demand — they differ only in whether they work around this limitation through discrete action abstraction or through external computational augmentation.

5 Connection to the Final Project

The **closest related work is FM-Planner** [Xiao et al., 2025], which establishes that LLM-generated sparse waypoint sequences are viable for drone navigation in structured environments and that text-coordinate representations consistently outperform VLM visual map inputs. The present project **extends this direction by introducing an explicit verification layer that evaluates generated waypoints against user-defined safety and feasibility metrics** and feeds failure diagnostics back into the prompt — a mechanism absent from FM-Planner and from most surveyed works. This directly addresses the **geometric imprecision and hallucination failure modes** identified across Cluster 1 and documented in the STMR spatial reasoning ablations [Gao et al., 2025].

The RAG-augmented hallucination mitigation strategy in LLM-Land [Cai et al., 2025] and the context-aware ICL framework from the ICL UAV paper [Emami et al., 2026] collectively motivate the design choice to **augment LLM prompts with structured environmental context and verification feedback rather than relying on zero-shot generation**. LLM-Land’s result that LLM-without-RAG achieves only 6 percent success rate while LLM-Land achieves 96 percent directly supports the hypothesis that ungrounded LLM outputs are insufficient for deployment and that structured external mechanisms are necessary to realize the semantic reasoning benefits of LLMs reliably. Though verification feedback couldn’t been incorporated in current codebase submitted but it can be a potential next step to extend this work

The **semantic encoding insights from the STMR paper** [Gao et al., 2025] motivate the use of **YOLO-based structured obstacle descriptions rather than raw point cloud data**. The **tradeoff this introduces — reduced generalizability in exchange for reliable semantic representations** — is consistent with the STMR finding that geometric-only or topological-only representations are significantly weaker than combined semantic-topological-metric representations.

SafePath [SafePath, 2025] illustrates the opposite end of the design space: by restricting LLM outputs to a discrete action vocabulary, it achieves safety guarantees but forfeits the semantic reasoning capability that constitutes the primary motivation for using LLMs in navigation. **The present project rejects this simplification in favor of coordinate-level waypoint generation augmented with post-hoc verification, preserving the full semantic expressiveness of LLM-based planning while addressing its numerical reliability limitations** through external mechanisms.

The LLM-augments-classical cluster — particularly SRAH [Durrani and Durrani, 2026] and LLM-DWA [LLM-DWA, 2025] — **validates the broader principle that FM contributions are most effective when they configure or guide classical systems rather than replace them.** The proposed trajectory refinement pipeline follows this same architectural principle: classical interpolation and PX4-based tracking retain responsibility for geometric feasibility and dynamic constraints, while the **LLM operates exclusively at the level of semantic route decisions.**

The gap this project addresses is the combination of verification-augmented replanning, **real hardware deployment on a VOXL 2 / Starling platform, and explicit architectural separation of semantic planning, geometric refinement, and safety verification into distinct stages.** The proposed framework is positioned as a synthesis of these contributions targeting **deployable, verification-grounded, semantically aware high-level route planning on real UAV hardware.**

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